

APPENDIX

This appendix provides additional experimental results, theoretical analysis, and algorithmic details. The content is organized as follows:

- **Appendix Section A:** Effects of Token Manipulation Ratio on Naïve Vision Token Manipulation Attacks
- **Appendix Section B:** Optimization-based Token Selection Algorithm
- **Appendix Section C:** Norm Bias in Self-Attention Disruption
- **Appendix Section D:** Adaptive λ Calibration

A. Effects of Token Manipulation Ratio on Naïve Vision Token Manipulation Attacks

In this section, we analyze how attack effectiveness changes as the TMR increases. We vary $\rho \in \{0.1, 0.2, \dots, 1.0\}$ and report the corresponding accuracy for each attack type and model in Tables II–V. Overall, a larger TMR generally leads to more severe degradation, but the rate of degradation differs substantially across attack primitives and model families. s

B. Algorithm

The detailed optimization procedure is summarized in Algorithm 1. During optimization, we construct the manipulated vision tokens using a differentiable relaxation of the token manipulation operator. For example, for sign flip, we adopt a continuous relaxation such that at iteration t , each token is transformed as

$$\tilde{v}_i^{(t)} = (1 - 2w_i^{(t-1)})v_i, \quad (8)$$

where $w_i^{(t-1)} \in [0, 1]$ controls the degree of transformation. This formulation provides a differentiable approximation to discrete sign flipping, where $w_i = 0$ keeps the original token unchanged and $w_i = 1$ corresponds to full sign inversion. The resulting manipulated token matrix $\tilde{V}^{(t)}$ is then used to evaluate the optimization objective.

C. Norm Bias in Self-Attention Disruption

We consider perturbations that modify a selected token either by value transformation or by token replacement. Formally, for each selected token v_i , the perturbed token is written as

$$\tilde{v}_i = \phi_i(V), \quad (9)$$

where ϕ_i denotes a transformation operator. For value-based attacks such as masking, Gaussian perturbation, and sign flipping, this reduces to the additive form $\tilde{v}_i = v_i + \delta_i$ with perturbation magnitude proportional to $\|v_i\|_2$. For structure-based attacks such as permutation, $\phi_i(V) = v_{\sigma(i)}$ for some derangement σ , and the resulting discrepancy depends on the pairwise token difference $\|v_{\sigma(i)} - v_i\|_2$, which is bounded by the overall token magnitude, e.g., $\|v_{\sigma(i)} - v_i\|_2 \leq 2\|V\|_F$.

Proof. Let $G(V) = VV^\top$ denote the pairwise similarity matrix of vision tokens. Suppose a selected token v_i is perturbed to

$$\tilde{v}_i = \phi_i(V) = v_i + \Delta_i, \quad (10)$$

Algorithm 1 Optimization-based Token Selection

Require: Vision tokens $V = [v_1, \dots, v_{L_v}]^\top$; token manipulation ratio ρ ; number of iterations T ; learning rate η ; regularization weight λ .

- 1: Set the manipulation budget: $k \leftarrow \lfloor \rho L_v \rfloor$.
- 2: Initialize the selection variables: $w_i^{(0)} \leftarrow \frac{k}{L_v}$, for all $i \in [1, L_v]$.
- 3: **for** $t \in [1, T]$ **do**
- 4: Construct the manipulated vision tokens $\tilde{V}^{(t)}$ from V and $w^{(t-1)}$.
- 5: Compute the self-attention disruption loss:

$$\mathcal{L}_{SA}(w^{(t-1)}) = \left\| \text{Attn}(\tilde{V}^{(t)}) - \text{Attn}(V) \right\|_F^2$$

- 6: Compute the norm-aware regularization:

$$\mathcal{L}_{\text{norm}}(w^{(t-1)}) = \sum_{i=1}^{L_v} w_i^{(t-1)} \hat{z}_i$$

- 7: Compute the objective:

$$\mathcal{L}(w^{(t-1)}) = \mathcal{L}_{SA}(w^{(t-1)}) - \lambda \mathcal{L}_{\text{norm}}(w^{(t-1)})$$

- 8: Update the selection variables:

$$w^{(t)} \leftarrow w^{(t-1)} + \eta \nabla_w \mathcal{L}(w^{(t-1)})$$

- 9: Clip $w^{(t)}$ element wise to $[0, 1]$.
- 10: Rescale $w^{(t)}$ to satisfy the budget constraint:

$$w^{(t)} \leftarrow w^{(t)} \cdot \frac{k}{\sum_{i=1}^{L_v} w_i^{(t)}}$$

- 11: Clip $w^{(t)}$ element wise to $[0, 1]$ again.
- 12: **end for**
- 13: Obtain the discrete token set: $\mathcal{I} \leftarrow \text{Top-k}(w^{(T)})$.
- 14: Apply the chosen token manipulation to the selected tokens in $\mathcal{I}$.
- 15: **return** $\mathcal{I}$

where

$$\Delta_i := \phi_i(V) - v_i \quad (11)$$

denotes the effective perturbation induced by the transformation operator ϕ_i . All other tokens remain unchanged.

Then, for any $j \neq i$, we have

$$\tilde{G}_{ij} - G_{ij} = \tilde{v}_i^\top v_j - v_i^\top v_j = (v_i + \Delta_i)^\top v_j - v_i^\top v_j = \Delta_i^\top v_j, \quad (12)$$

and similarly

$$\tilde{G}_{ji} - G_{ji} = \Delta_i^\top v_j. \quad (13)$$

For the diagonal entry,

$$\tilde{G}_{ii} - G_{ii} = \|\tilde{v}_i\|_2^2 - \|v_i\|_2^2 = \|v_i + \Delta_i\|_2^2 - \|v_i\|_2^2 = 2v_i^\top \Delta_i + \|\Delta_i\|_2^2. \quad (14)$$

All other entries remain unchanged. Therefore,

$$\|\tilde{G} - G\|_F^2 = 2 \sum_{j \neq i} (\Delta_i^\top v_j)^2 + (2v_i^\top \Delta_i + \|\Delta_i\|_2^2)^2. \quad (15)$$

TABLE II
PERFORMANCE UNDER DIFFERENT TOKEN MANIPULATION ATTACKS WITH VARYING TMR ON OCRBENCH.

Model	Benign	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Permutation											
Qwen2.5-VL-3B-Instruct	78.90	77.60	75.80	74.90	73.30	72.70	70.60	71.10	69.20	70.00	70.20
Qwen2.5-VL-7B-Instruct	83.70	83.10	82.20	82.40	80.60	80.40	79.60	78.40	77.50	77.40	77.70
Qwen2.5-VL-72B-Instruct	84.00	83.50	82.60	81.70	81.20	79.80	79.20	78.10	78.40	78.90	78.70
InternVL3.5-4B	81.30	79.90	76.70	72.40	68.20	61.20	50.30	39.60	30.80	24.80	22.60
InternVL3.5-8B	82.70	80.90	76.50	72.50	66.20	57.60	45.10	36.30	29.80	24.70	22.40
InternVL3.5-38B	86.60	85.80	82.40	78.20	73.60	65.70	55.70	45.30	35.20	28.90	26.40
Masking											
Qwen2.5-VL-3B-Instruct	78.90	73.50	64.90	58.70	52.80	45.90	39.90	29.70	21.10	9.90	0.10
Qwen2.5-VL-7B-Instruct	83.70	75.00	61.20	45.50	22.70	4.60	0.20	0.00	0.00	0.00	0.20
Qwen2.5-VL-72B-Instruct	84.00	82.90	78.10	73.00	69.10	65.30	61.80	55.60	45.20	28.00	2.40
InternVL3.5-4B	81.30	79.70	76.90	75.00	71.20	65.20	56.80	45.80	31.40	12.20	1.80
InternVL3.5-8B	82.70	81.70	78.50	75.30	71.30	60.80	50.90	37.40	21.60	9.80	1.40
InternVL3.5-38B	86.60	85.70	83.60	82.20	80.20	76.50	70.00	58.50	45.10	25.40	2.10
Gaussian Perturbation											
Qwen2.5-VL-3B-Instruct	78.90	66.80	57.20	51.40	45.90	38.60	33.00	27.20	21.30	11.40	0.10
Qwen2.5-VL-7B-Instruct	83.70	75.50	65.40	57.80	50.50	44.40	38.30	35.50	27.80	17.70	0.10
Qwen2.5-VL-72B-Instruct	84.00	16.40	18.40	15.80	20.00	18.40	15.50	12.50	10.00	5.60	0.00
InternVL3.5-4B	81.30	79.00	74.20	71.70	63.30	57.40	46.40	37.40	24.50	10.80	1.00
InternVL3.5-8B	82.70	79.30	71.50	58.40	42.90	29.60	19.30	11.70	5.80	2.80	0.80
InternVL3.5-38B	86.60	84.20	82.90	80.40	76.40	71.00	63.90	50.50	38.00	17.10	1.70
Sign Flip											
Qwen2.5-VL-3B-Instruct	78.90	35.70	23.10	15.10	7.80	3.90	1.70	0.40	0.20	0.00	0.10
Qwen2.5-VL-7B-Instruct	83.70	19.80	9.90	6.10	2.70	1.50	1.30	0.90	0.30	0.00	0.00
Qwen2.5-VL-72B-Instruct	84.00	2.00	3.00	2.50	3.30	2.10	2.50	2.10	0.90	0.40	0.00
InternVL3.5-4B	81.30	18.00	11.50	8.50	4.90	3.80	2.70	2.10	1.30	0.30	0.30
InternVL3.5-8B	82.70	68.60	56.70	48.50	40.00	31.30	24.00	15.40	7.80	3.40	0.80
InternVL3.5-38B	86.60	82.30	76.50	72.70	65.60	58.30	48.60	37.10	25.60	11.90	0.60

Using the Cauchy–Schwarz inequality,

$$(\Delta_i^\top v_j)^2 \leq \|\Delta_i\|_2^2 \|v_j\|_2^2, \quad (16)$$

which implies

$$\|\tilde{G} - G\|_F^2 \leq 2\|\Delta_i\|_2^2 \sum_{j \neq i} \|v_j\|_2^2 + (2v_i^\top \Delta_i + \|\Delta_i\|_2^2)^2. \quad (17)$$

This shows that the change in the similarity matrix is governed by the effective perturbation magnitude $\|\Delta_i\|_2$. Under the assumption that the induced discrepancy is controlled by the token magnitude or, more generally, bounded by the overall token energy, tokens associated with larger effective perturbations tend to induce larger changes in $VV^\top$.

Since the self-attention operator is a smooth function of $VV^\top$, larger perturbations to $VV^\top$ generally lead to larger deviations in the resulting attention patterns. Consequently, when maximizing the self-attention disruption objective $\mathcal{L}_{\text{SA}}$, the optimization tends to favor tokens whose transformation induces larger effective discrepancies in the similarity structure.

D. Adaptive λ Calibration

The relative scales of $\mathcal{L}_{\text{SA}}$ and $\mathcal{L}_{\text{norm}}$ vary across inputs due to differences in vision token distributions. A fixed λ cannot maintain a consistent balance between the two terms. We therefore calibrate λ adaptively. Given the uniform initialization $w_i^{(0)} = \frac{k}{L_v}$, we perform a single forward-backward pass to estimate the average gradient magnitudes:

$$\bar{g}_{\text{SA}} = \text{mean}(|\nabla_w \mathcal{L}_{\text{SA}}|), \quad \bar{g}_{\text{norm}} = \text{mean}(|\nabla_w \mathcal{L}_{\text{norm}}|). \quad (18)$$

We then set

$$\lambda = \alpha \cdot \frac{\bar{g}_{\text{SA}}}{\bar{g}_{\text{norm}}}, \quad (19)$$

with $\alpha = 0.5$. By setting λ proportional to $\bar{g}_{\text{SA}}/\bar{g}_{\text{norm}}$, the gradient magnitude of the regularization term is adjusted to be comparable to that of the self-attention disruption term at the starting point. The coefficient α further controls the relative strength of the regularization, allowing $\mathcal{L}_{\text{SA}}$ to remain the primary objective while preventing the optimization from being overly biased toward high-norm tokens.

TABLE III
PERFORMANCE UNDER DIFFERENT TOKEN MANIPULATION ATTACKS WITH VARYING TMR ON MMBENCH.

Model	Benign	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Permutation											
Qwen2.5-VL-3B-Instruct	76.54	76.39	76.39	76.00	75.07	75.23	74.53	74.61	75.15	73.91	73.99
Qwen2.5-VL-7B-Instruct	79.87	79.56	79.25	79.02	78.86	78.63	78.56	78.09	78.09	78.01	77.39
Qwen2.5-VL-72B-Instruct	88.39	87.77	88.23	87.30	87.07	87.22	86.91	86.84	86.14	86.60	86.14
InternVL3.5-4B	81.03	79.56	78.71	77.63	76.62	74.76	73.52	72.21	71.28	70.66	70.12
InternVL3.5-8B	83.20	81.03	79.56	78.17	76.39	75.77	72.98	72.91	71.28	69.89	69.73
InternVL3.5-38B	86.99	85.75	84.21	82.19	81.42	80.41	79.64	78.17	76.85	77.01	75.77
Masking											
Qwen2.5-VL-3B-Instruct	76.54	74.14	71.82	70.89	67.49	61.14	40.55	2.47	0.00	0.00	0.00
Qwen2.5-VL-7B-Instruct	79.87	24.69	7.35	2.47	0.61	0.07	0.07	0.00	0.00	0.00	0.00
Qwen2.5-VL-72B-Instruct	88.39	88.31	87.92	87.69	86.37	81.81	75.15	76.70	80.65	70.89	13.31
InternVL3.5-4B	81.03	80.03	79.41	78.79	77.78	76.00	74.53	73.06	70.20	60.91	17.87
InternVL3.5-8B	83.20	81.42	81.26	79.64	78.79	77.55	77.01	73.99	69.42	61.22	16.40
InternVL3.5-38B	86.99	86.53	85.99	85.91	84.98	84.75	83.43	82.04	79.48	73.45	19.42
Gaussian Perturbation											
Qwen2.5-VL-3B-Instruct	76.54	49.77	43.89	38.78	31.89	26.63	17.26	11.15	5.34	2.17	1.01
Qwen2.5-VL-7B-Instruct	79.87	62.46	64.55	54.95	48.68	37.69	25.69	16.79	1.70	0.85	0.46
Qwen2.5-VL-72B-Instruct	88.39	32.43	5.88	0.85	0.30	0.30	0.30	0.07	0.23	0.00	0.00
InternVL3.5-4B	81.03	78.25	77.48	76.24	74.07	73.30	69.89	68.19	63.70	54.18	13.31
InternVL3.5-8B	83.20	80.11	77.94	75.08	68.42	62.54	51.63	48.60	41.02	28.79	6.03
InternVL3.5-38B	86.99	86.15	85.45	85.60	84.52	84.37	82.66	80.11	76.24	68.96	16.18
Sign Flip											
Qwen2.5-VL-3B-Instruct	76.54	2.01	1.23	1.00	0.77	0.38	0.46	0.30	0.23	0.15	0.07
Qwen2.5-VL-7B-Instruct	79.87	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Qwen2.5-VL-72B-Instruct	88.39	0.23	0.23	0.38	0.23	0.23	0.30	0.23	0.23	0.23	0.30
InternVL3.5-4B	81.03	27.24	22.67	19.65	18.26	16.95	16.25	14.24	12.46	9.52	1.70
InternVL3.5-8B	83.20	79.72	76.93	73.07	68.96	63.62	56.97	49.07	44.73	29.64	8.04
InternVL3.5-38B	86.99	86.46	85.76	84.60	81.42	74.85	69.12	59.83	56.57	47.75	10.60

TABLE IV
PERFORMANCE UNDER DIFFERENT TOKEN MANIPULATION ATTACKS WITH VARYING TMR ON HALLUSIONBENCH.

Model	Benign	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Permutation											
Qwen2.5-VL-3B-Instruct	60.04	59.31	58.99	60.25	58.89	58.25	57.94	58.15	56.78	57.31	57.41
Qwen2.5-VL-7B-Instruct	65.19	64.25	63.51	63.41	62.04	62.15	61.93	61.72	62.15	61.93	61.41
Qwen2.5-VL-72B-Instruct	69.61	68.98	68.03	68.03	67.61	66.56	66.25	65.51	65.09	64.25	64.77
InternVL3.5-4B	69.08	68.24	67.61	67.40	67.19	64.98	65.72	61.72	60.78	60.15	60.15
InternVL3.5-8B	71.39	69.40	68.45	66.25	64.98	64.98	62.25	59.52	58.25	57.31	58.15
InternVL3.5-38B	75.07	73.19	72.13	72.24	68.87	68.66	66.04	62.46	61.83	60.67	60.88
Masking											
Qwen2.5-VL-3B-Instruct	60.04	57.41	55.10	51.74	46.69	38.38	26.29	7.47	1.26	0.74	0.11
Qwen2.5-VL-7B-Instruct	65.19	29.34	16.51	4.21	0.63	0.00	0.00	0.00	0.00	0.00	0.00
Qwen2.5-VL-72B-Instruct	69.61	68.56	68.03	65.62	65.19	63.20	65.72	62.67	60.99	52.89	46.37
InternVL3.5-4B	69.08	68.66	68.77	67.09	67.09	64.88	64.98	63.72	62.46	56.15	54.99
InternVL3.5-8B	71.39	71.82	69.93	70.45	68.35	65.72	63.62	62.25	59.20	54.99	53.10
InternVL3.5-38B	75.07	74.66	74.13	73.08	72.56	70.45	70.66	67.93	65.62	60.67	54.26
Gaussian Perturbation											
Qwen2.5-VL-3B-Instruct	60.04	45.53	38.49	28.92	20.08	13.88	10.41	6.62	5.36	4.73	4.42
Qwen2.5-VL-7B-Instruct	65.19	59.41	51.74	38.07	17.25	10.30	6.62	3.15	2.63	2.10	1.05
Qwen2.5-VL-72B-Instruct	69.61	22.61	6.52	0.32	0.21	0.11	0.00	0.00	0.00	0.00	0.00
InternVL3.5-4B	69.08	67.72	67.30	67.72	64.98	64.04	62.46	63.20	58.68	57.20	55.10
InternVL3.5-8B	71.39	68.66	68.35	65.83	62.88	59.73	53.84	49.63	40.69	29.34	18.19
InternVL3.5-38B	75.07	73.40	73.40	71.50	71.40	68.77	69.09	64.04	59.83	55.52	51.84
Sign Flip											
Qwen2.5-VL-3B-Instruct	60.04	0.00	0.21	0.00	0.00	0.21	0.32	0.11	0.00	0.00	0.00
Qwen2.5-VL-7B-Instruct	65.19	0.00	0.00	0.00	0.00	0.00	0.00	0.11	0.00	0.00	0.00
Qwen2.5-VL-72B-Instruct	69.61	0.11	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
InternVL3.5-4B	69.08	36.07	23.66	18.09	14.72	14.20	12.30	10.52	10.62	10.83	9.36
InternVL3.5-8B	71.39	68.98	67.61	66.77	63.72	60.36	59.20	53.52	51.95	45.95	33.12
InternVL3.5-38B	75.07	73.40	72.77	71.50	70.56	66.98	64.56	60.15	54.68	52.16	46.79

TABLE V
PERFORMANCE UNDER DIFFERENT TOKEN MANIPULATION ATTACKS WITH VARYING TMR ON MATHVISTA.

Model	Benign	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
Permutation											
Qwen2.5-VL-3B-Instruct	56.40	57.10	56.40	56.90	56.60	55.10	54.90	55.40	54.30	52.90	52.30
Qwen2.5-VL-7B-Instruct	45.40	44.20	44.20	44.80	45.30	42.90	44.30	44.40	44.40	43.90	42.00
Qwen2.5-VL-72B-Instruct	47.30	27.90	19.30	20.05	22.80	19.60	19.50	24.10	27.60	22.80	26.20
InternVL3.5-4B	49.90	48.10	45.20	45.10	41.70	40.40	36.30	34.90	34.20	32.00	32.90
InternVL3.5-8B	48.70	47.80	46.50	44.80	40.80	37.80	35.90	34.20	33.80	32.70	31.50
InternVL3.5-38B	56.60	52.50	51.10	50.40	46.60	43.10	41.40	38.90	35.90	36.00	35.80
Masking											
Qwen2.5-VL-3B-Instruct	56.40	54.10	50.40	46.50	42.50	37.10	33.00	25.20	20.30	19.20	16.60
Qwen2.5-VL-7B-Instruct	45.40	33.80	31.00	24.80	19.90	16.20	16.80	17.70	18.70	18.00	17.20
Qwen2.5-VL-72B-Instruct	47.30	36.40	37.90	35.80	42.50	38.50	28.60	35.20	31.40	23.70	15.80
InternVL3.5-4B	49.90	51.80	49.30	49.70	47.60	47.60	44.40	41.20	37.30	30.40	23.20
InternVL3.5-8B	48.70	49.40	48.70	48.70	47.90	47.60	44.20	41.40	38.50	36.00	26.30
InternVL3.5-38B	56.60	56.30	56.49	54.90	55.20	52.50	48.69	45.40	42.19	35.30	28.90
Gaussian Perturbation											
Qwen2.5-VL-3B-Instruct	56.40	43.80	36.80	35.60	30.50	27.30	25.10	24.60	21.90	20.50	20.40
Qwen2.5-VL-7B-Instruct	45.40	42.50	37.40	32.50	26.00	21.80	21.20	23.20	20.40	21.90	23.10
Qwen2.5-VL-72B-Instruct	47.30	31.80	25.50	21.60	21.50	20.40	22.10	18.90	20.50	19.00	19.10
InternVL3.5-4B	49.90	50.60	49.40	50.40	46.40	42.90	39.70	35.10	34.20	29.00	25.20
InternVL3.5-8B	48.70	42.90	38.30	35.90	34.00	30.60	27.70	24.40	22.90	21.00	19.30
InternVL3.5-38B	56.60	54.90	55.20	52.50	54.30	50.60	47.90	45.00	41.09	36.90	30.50
Sign Flip											
Qwen2.5-VL-3B-Instruct	56.40	19.10	19.30	20.80	19.40	18.00	17.80	18.40	18.50	18.30	16.30
Qwen2.5-VL-7B-Instruct	45.40	16.80	15.60	15.20	15.90	16.10	15.70	15.00	15.30	14.90	16.80
Qwen2.5-VL-72B-Instruct	47.30	16.70	16.70	18.00	17.40	17.29	15.29	16.70	16.40	15.80	18.20
InternVL3.5-4B	49.90	38.00	32.30	30.70	29.70	27.50	25.40	23.40	23.10	21.30	19.70
InternVL3.5-8B	48.70	43.70	41.90	40.00	39.60	37.90	34.20	31.00	27.90	23.80	20.50
InternVL3.5-38B	56.60	50.50	51.50	50.70	50.70	50.00	46.80	45.40	42.40	35.50	27.70